

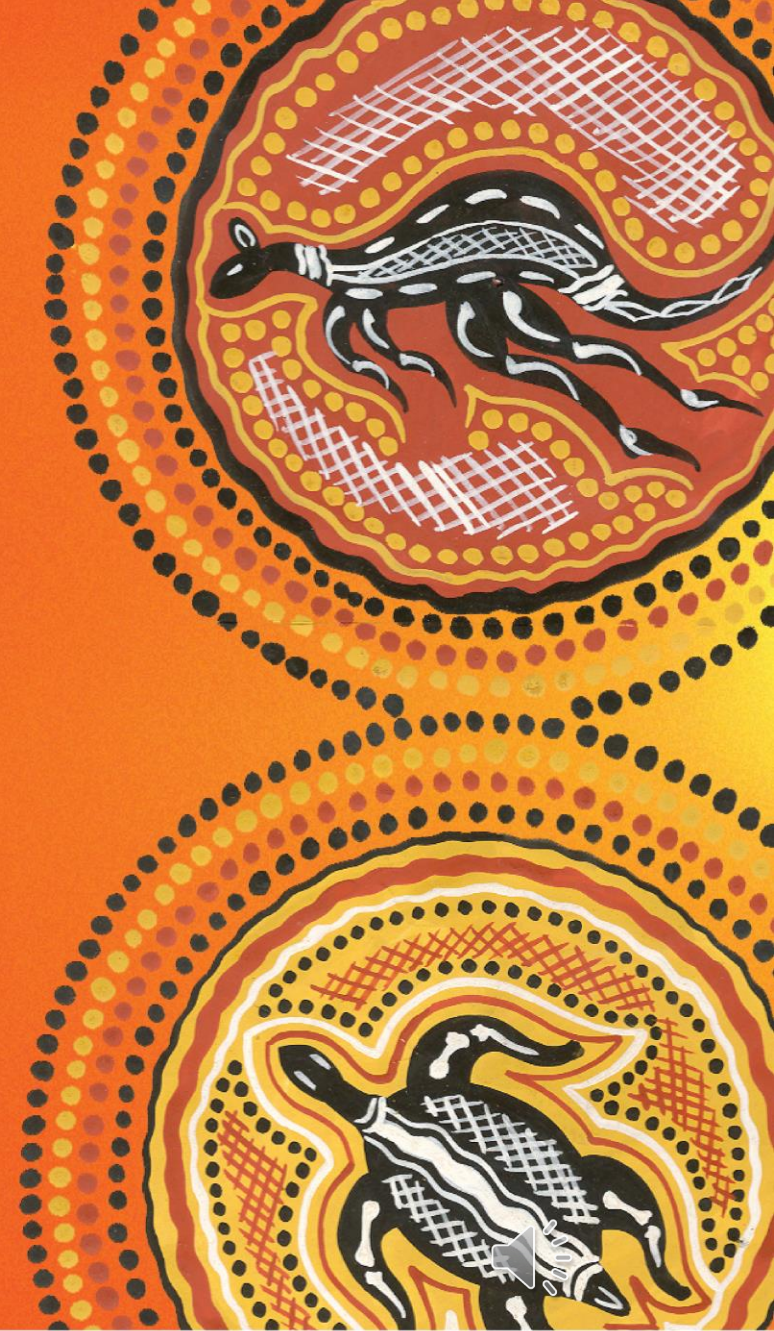
ICT702 Data Visualisation

Telling the Truth with Data Visualisation

Module 9



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



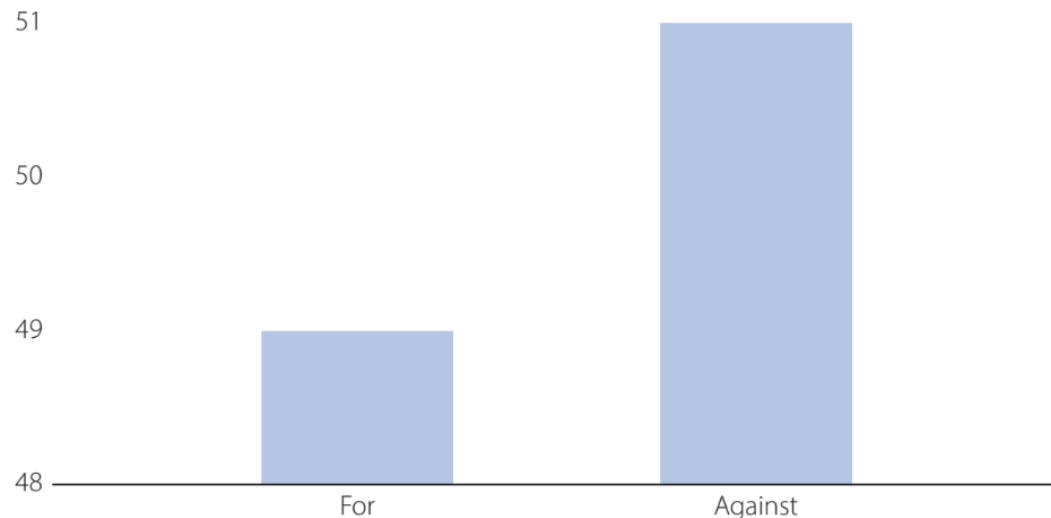
Part 3

Improper Chart Axes: Exaggerated Difference

Column chart that exaggerates difference in proportion of likely voters

Majority of Voters Against Library Levy

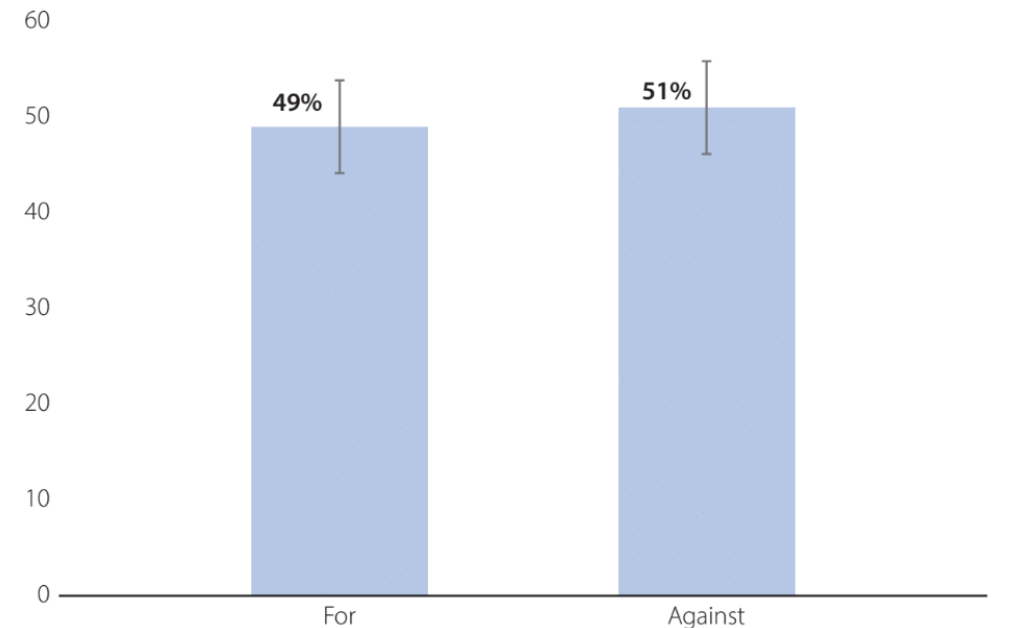
Percent (%) of Likely Voters



Redesigned column chart showing just a small difference in proportion of likely voters

Fate of Library Levy Uncertain

Percent (%) of Likely Voters

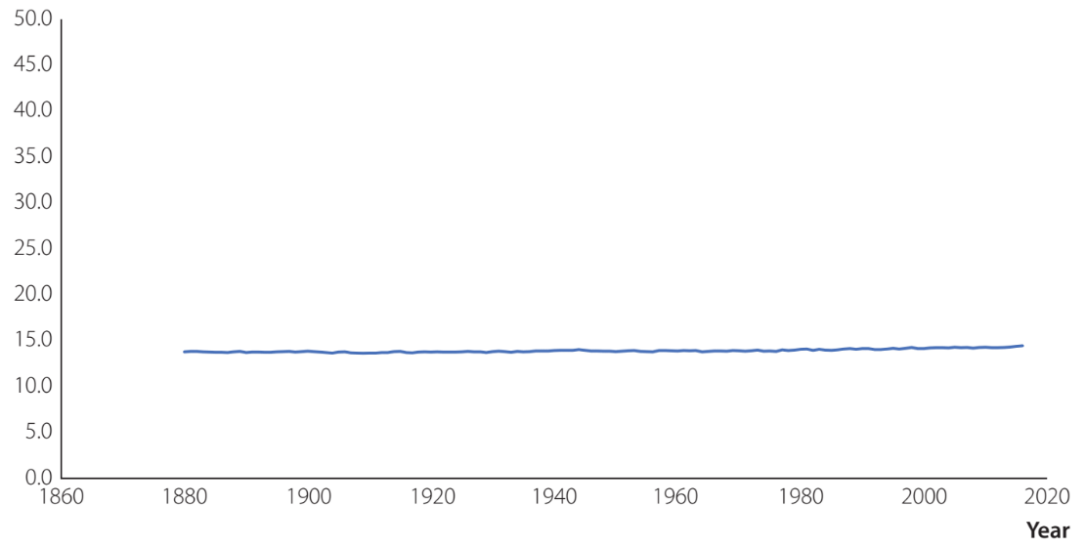


Improper Chart Axes: Hidden Insight

A line chart with a wide range showing no increase in the earth's surface temperature

Average Annual Global Surface Temperature

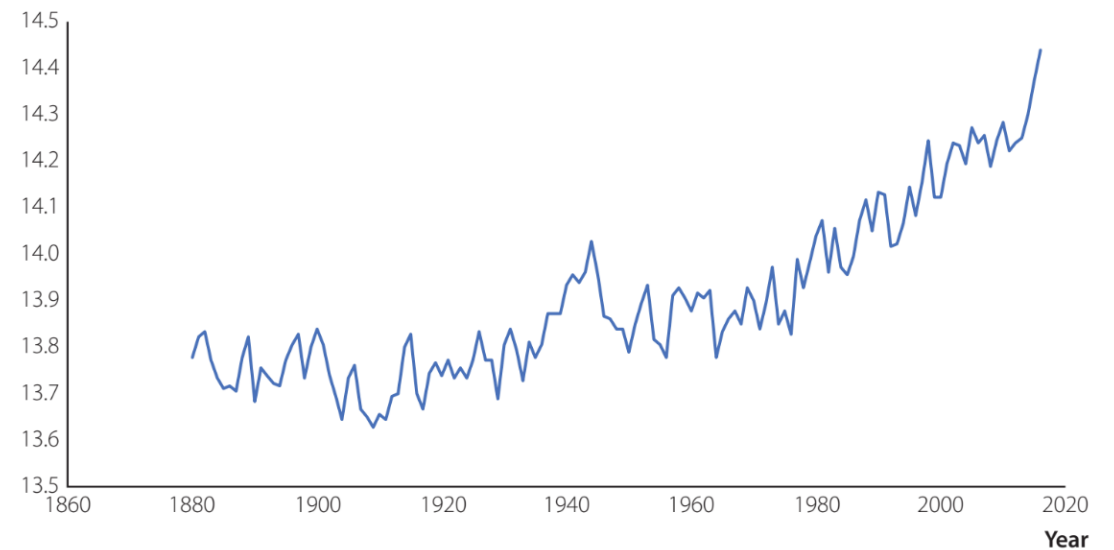
Temperature (°C)



A line chart with a revised range showing an increase in the earth's surface temperature

Average Annual Global Surface Temperature

Temperature (°C)

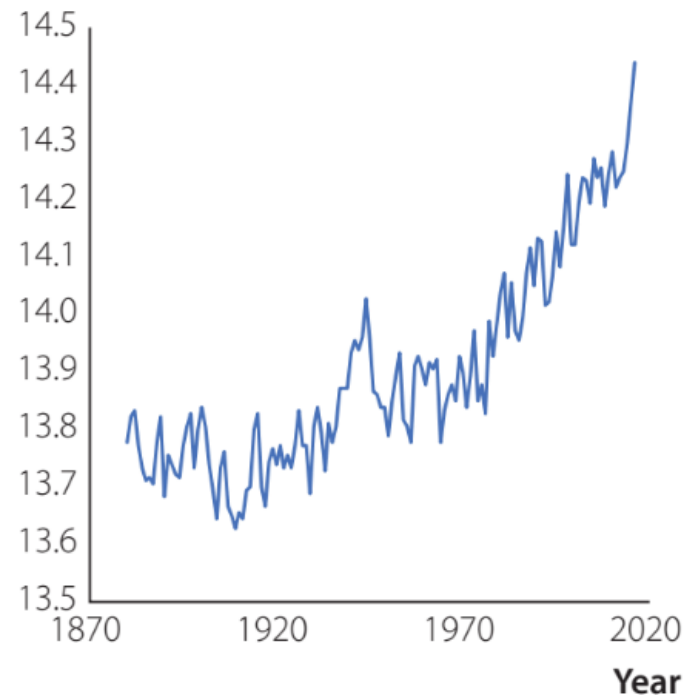


Improper Chart Axes: Incorrect Aspect Ratio

Two different line charts for average annual global surface temperature data showing effect of line chart aspect ratio on insights conveyed to audience

Average Annual Global Surface Temperature

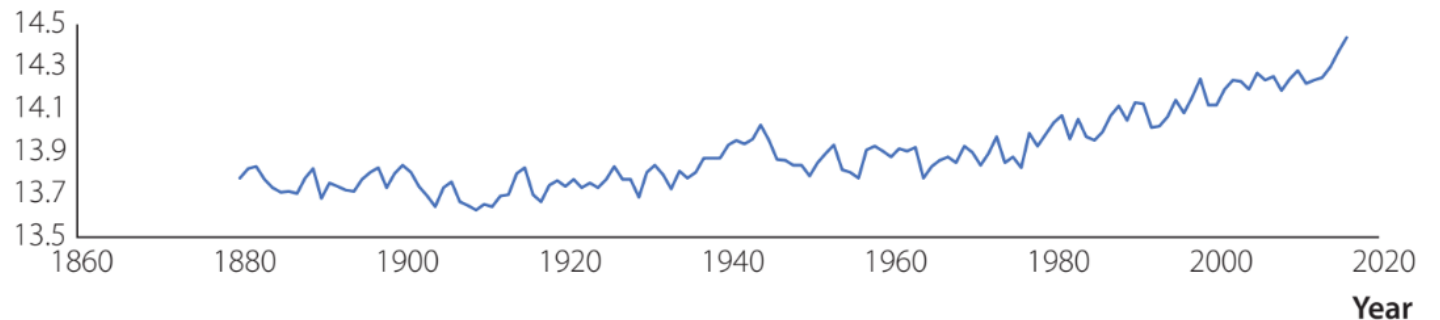
Temperature (°C)



(a)

Average Annual Global Surface Temperature

Temperature (°C)



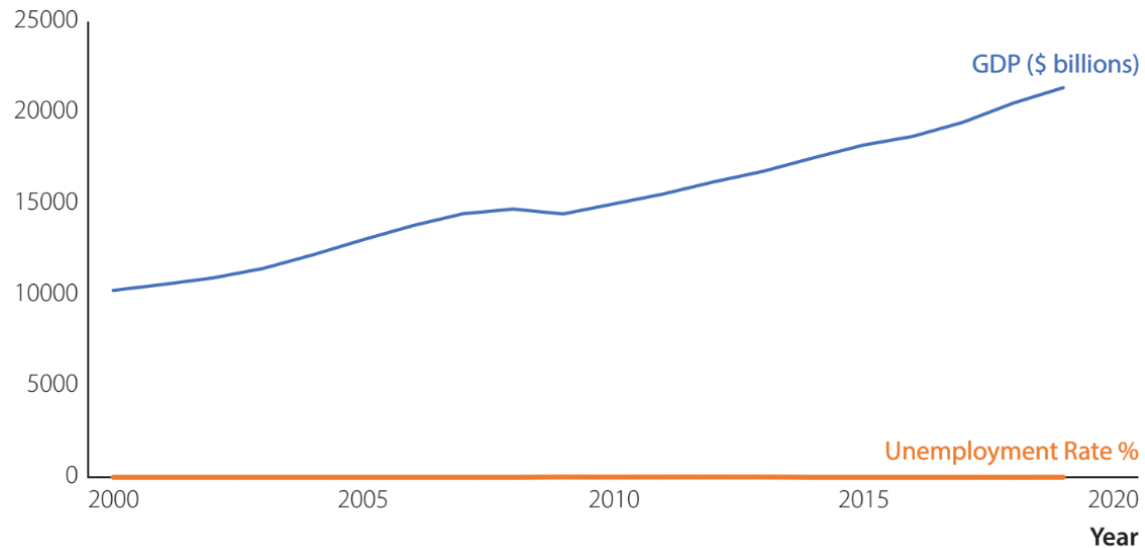
(b)

The aspect ratio refers to the width to height ratio of a chart.

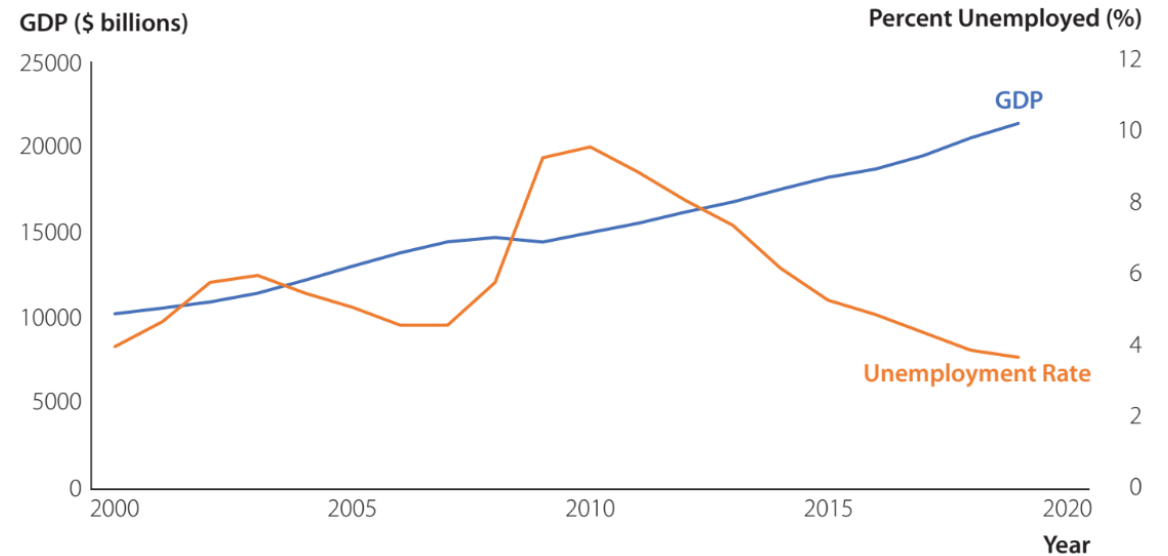
Dual-Axis Charts: Deceptive Design

Effect of using a dual-axis chart to display the gross domestic product and unemployment rate in the United States since 2000

Comparing Gross Domestic Product and Unemployment Rate in the United States Since 2000



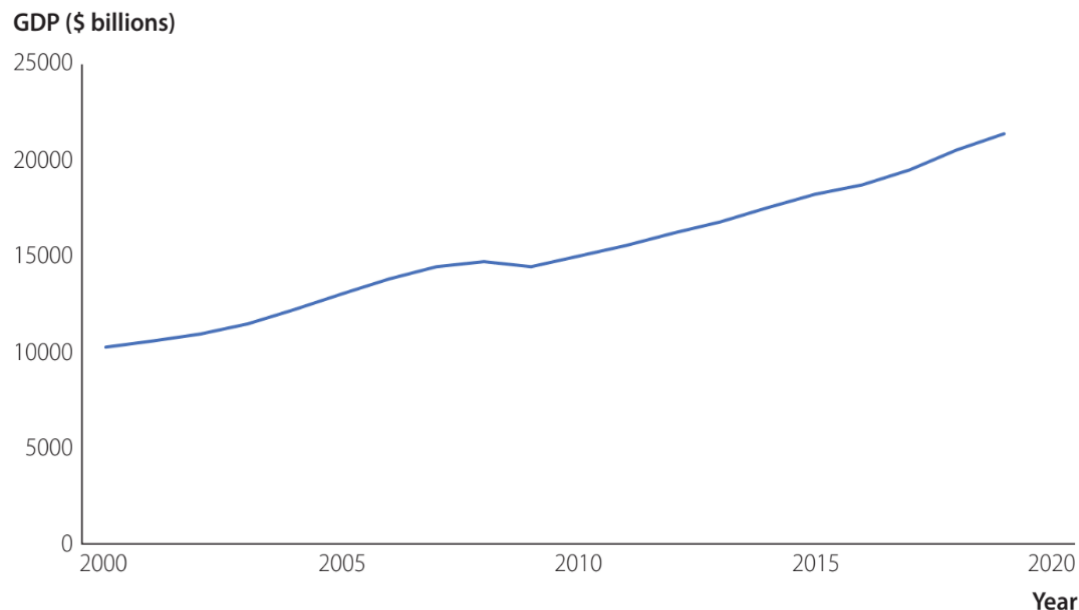
Comparing Gross Domestic Product and Unemployment Rate in the United States Since 2000



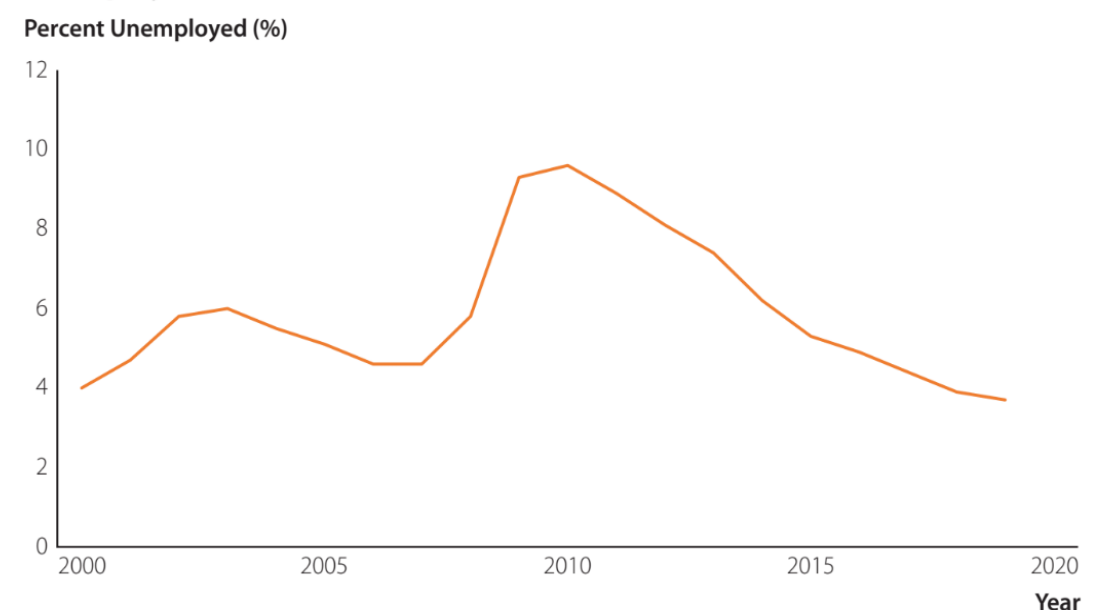
Dual-Axis Charts: Remedy

Using two different charts to display gross domestic product and unemployment rate in the US since 2000

Gross Domestic Product in the United States Since 2000



Unemployment Rate in the United States Since 2000



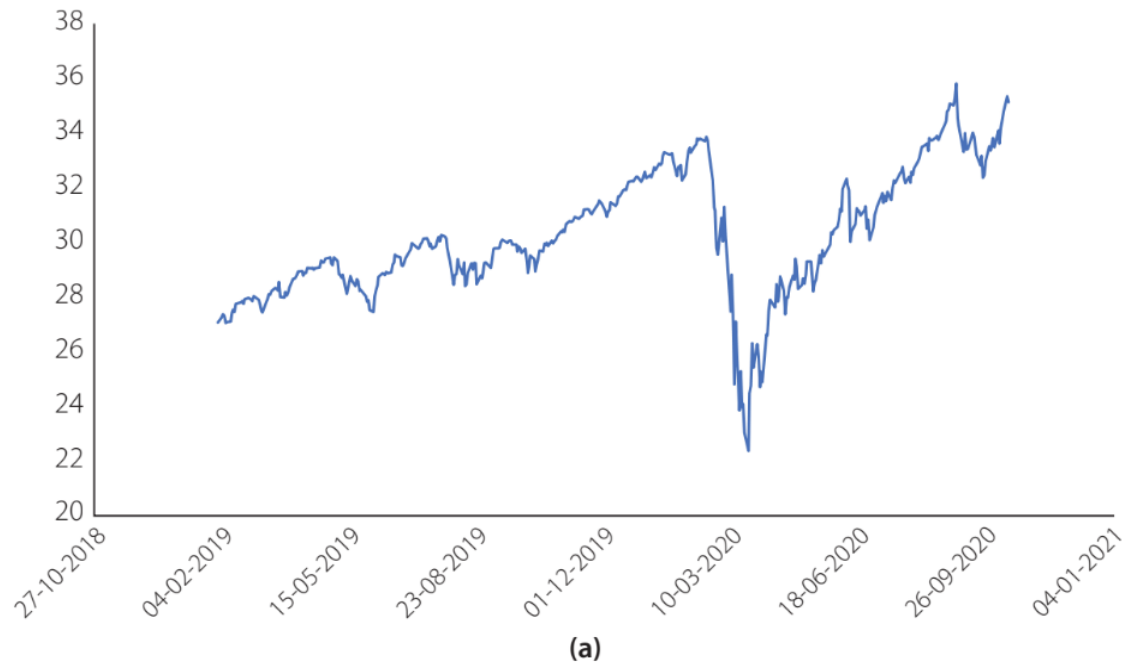
A simpler alternative to a dual-axis chart consists of using two separate charts with each variable shown individually.

Data Selection and Temporal Frequency

Comparing stock prices with same start and end dates but different temporal frequencies

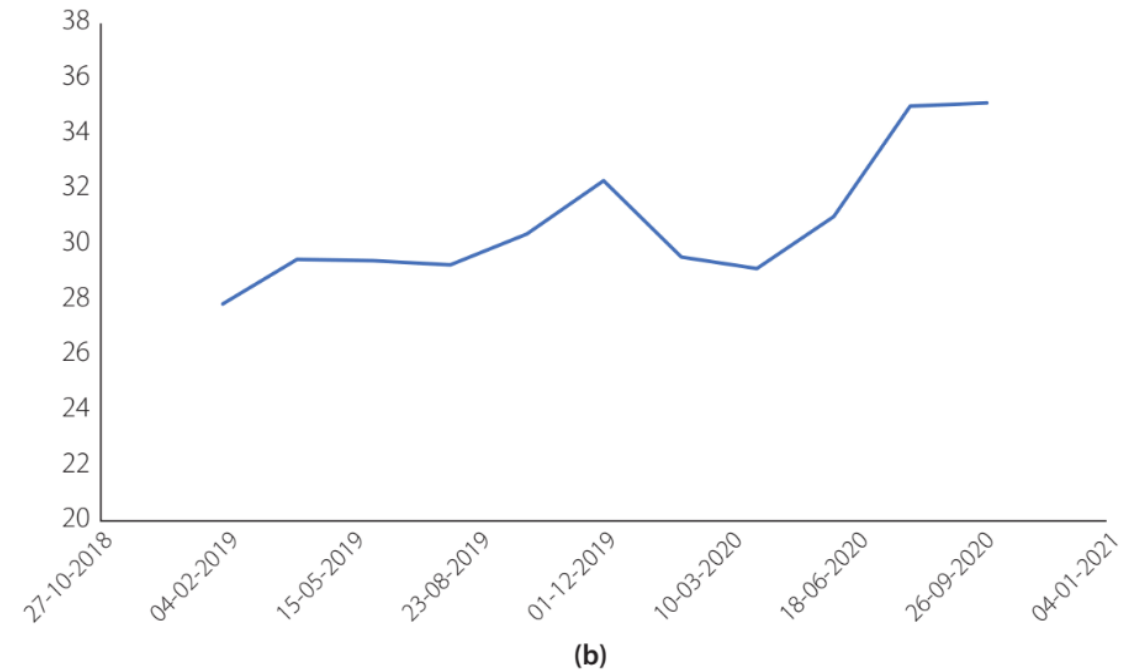
Stock Values Shown with Temporal Frequency = 365 per Year

Share Price (\$)



Stock Values Shown with Temporal Frequency = 6 per Year

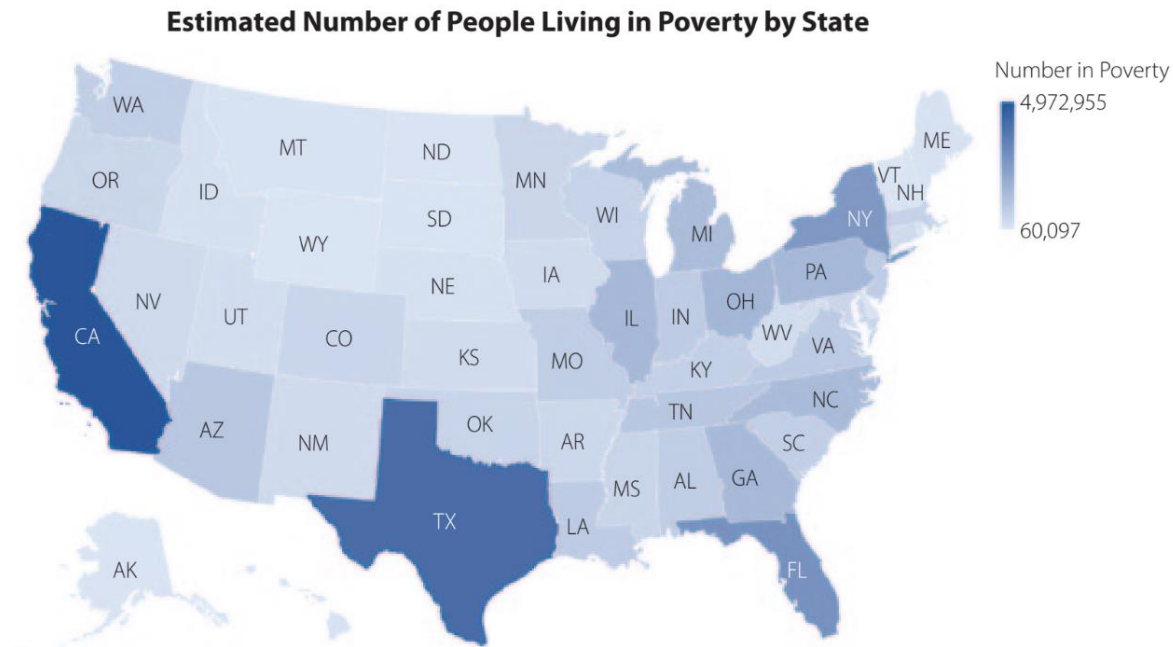
Share Price (\$)



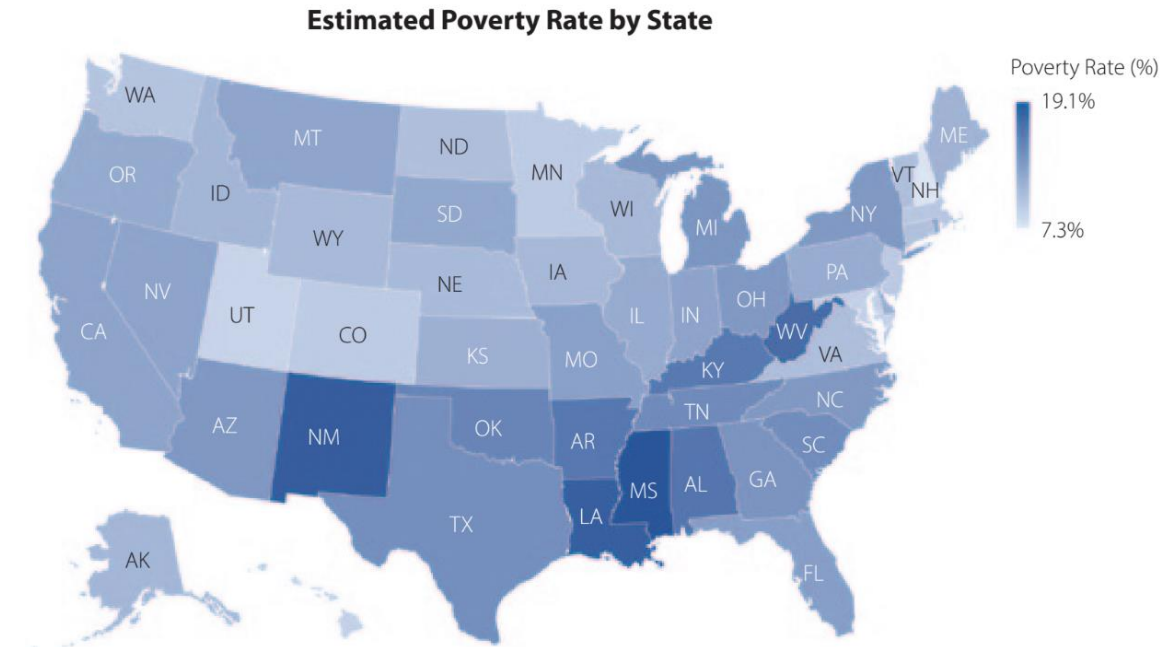
Viewing time series data at different temporal frequencies can reveal different patterns.

Issues Related to Geographic Maps

Choropleth map showing number of people living in poverty by state



Choropleth map showing poverty rate by state



Poverty rate by state is a better choropleth map for communicating which states have highest number of people living in poverty relative to overall state population.

Summary

In Module 9 you have learnt how:

- To identify missing data and data errors using different Excel methods that depend on the nature of the type of data and problem settings being analysed.
- Biased data may mislead insight from data visualisation and how to identify it.
- To adjust for inflation when dealing with time series data that have been collected over long time periods.
- Different ranges of values for the axes can greatly affect the insights conveyed to the audience for the same data.
- To remedy dual-axis charts that may often be confusing or misleading for the audience.
- The selection and frequency of time series data can change the insights conveyed to the audience.
- To prevent misleading the audience when using geographic maps.

End of Part 3

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